Curriculum Vitae of Kalins Banerjee

Personal Information

Country of Citizenship: India
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Research Interest

My research lies at the intersection of biostatistics, data science, spatial omics, and microbiome studies. I develop data-adaptive statistical and deep learning methods that transform high-dimensional complex biomedical data into clinically meaningful biological insights. I plan to continue building interdisciplinary collaborations that motivate new methodological development while generating insights into diverse biomedical problems.

Academic Experience

- Research Investigator, Research Faculty Track (starting September, 2026)
  Department of Urology, University of Michigan, USA.
  Mentor: Professor Evan T. Keller
  Research area: spatial multi-omics, spatial biology in 3D, and cancer biology.
- Research Fellow (July, 2025 - August, 2026)
  Department of Urology, University of Michigan, USA.
  Adviser: Professor Evan T. Keller
  Research area: single-cell, spatial transcriptomics, and cancer biology.
- Postdoctoral Fellow (September, 2021 - June, 2025)
  Department of Biostatistics, University of Michigan, USA.
  Adviser: Professor Xiang Zhou
  Research area: spatial transcriptomics, copy number variations, and intratumor heterogeneity.

Education

- Ph.D. in Biostatistics (2021), Pennsylvania State University, USA.
  Adviser: Professor Xiang Zhan
  Dissertation title: Adaptive Statistical Methods for Microbiome Association Analysis [Link].
- Master of Statistics (2017), Indian Statistical Institute, India.
  Specialization: Applied Statistics.
- Bachelor of Science (2015) with Honours in Statistics, Asutosh College, University of Calcutta, India.

Research Articles [Google Scholar]

Preprint

1. **Banerjee, K.**, Langefeld, R. C., Keller, E. T., & Zhou, X. (2026). SPICE: A Robust Computational Framework for Identifying Copy Number Variations in Spatial Transcriptomics. ***bioRxiv***, 2026-06. *Under review in Genome Medicine*. [Paper; R software]

Published

2. Olson, A. W., Li, J., Li, X. Y., King, L., Jiang, L., **Banerjee, K.**, ... & Weiss, S. J. (2026). Amoeboid–mesenchymal transition and the proteolytic control of cancer invasion plasticity. ***Proceedings of the National Academy of Sciences***, 123(15), e2520717123. [Paper; Scopus Q: $\mathbf{Q}_1$]

3. Poulsen, A., Jang, D. H., Khan, M., Al-Mohtaseb, Z. N., Chen, M., **Banerjee, K.**, Scott, I. U., & Pantanelli, S. M. (2025). Repeatability of a Dual-Scheimpflug Placido Disc Corneal Tomographer/Topographer in Eyes with Keratoconus. ***Clinical Ophthalmology***, 19, 2751–2758. [Paper; Scopus Q: $\mathbf{Q}_2$]

4. Nolan, Z. T., **Banerjee, K.**, Cong, Z., Gettle, S. L., Longenecker, A. L., Kawasawa, Y. I., ... & Nelson, A. M. (2023). Treatment response to isotretinoin correlates with specific shifts in Cutibacterium acnes strain composition within the follicular microbiome. ***Experimental Dermatology***, 32(7), 955-964. [Paper; Scopus Q: $\mathbf{Q}_1$]

5. Schneider, A. M., Nolan, Z. T., **Banerjee, K.**, Paine, A. R., Cong, Z., Gettle, S. L., ... & Nelson, A. M. (2023). Evolution of the facial skin microbiome during puberty in normal and acne skin. ***Journal of the European Academy of Dermatology and Venereology***, 37(1), 166-175. [Paper; Scopus Q: $\mathbf{Q}_1$]

6. **Banerjee, K.**, Chen, J., & Zhan, X. (2022). Adaptive and powerful microbiome multivariate association analysis via feature selection. ***Nucleic Acids Research (NAR) Genomics and Bioinformatics***, 4(1), lqab120. [Paper; R software; Scopus Q: $\mathbf{Q}_1$]

   [An earlier version won ***JSM (Joint Statistical Meetings) 2021 Paper Award*** **from the Biometrics section of the American Statistical Association (ASA)** (Announcement) (Penn State News) and ***Best Student Paper Award in Applied Statistics*** **at the IISA (International Indian Statistical Association) 2021 conference.**]

7. Zhan, X., **Banerjee, K.**, & Chen, J. (2021). Variant-Set Association Test for Generalized Linear Mixed Model. ***Genetic Epidemiology***, 45(4), 402-412. [Paper; R software; Scopus Q: $\mathbf{Q}_3$]

8. Kansara, N., Cui, D., **Banerjee, K.**, Landis, Z., Scott, I. U., & Pantanelli, S. M. (2021). Anterior, posterior, and nonkeratometric contributions to refractive astigmatism in pseudophakes. ***Journal of Cataract & Refractive Surgery***, 47(1), 93-99. [Paper; Scopus Q: $\mathbf{Q}_1$]

9. Ruzieh, M., Rogers, A. M., **Banerjee, K.**, Soleymani, T., Zhan, X., Foy, A. J., & Peterson, B. R. (2020). Safety of Bariatric Surgery in Patients with Coronary Artery Disease. ***Surgery for Obesity and Related Diseases***, 16(12), 2031-2037. [Paper; Scopus Q: $\mathbf{Q}_1$]

10. Schneider, A. M., Cook, L. C., Zhan, X., **Banerjee, K.**, Cong, Z., Imamura-Kawasawa, Y., ... & Nelson, A. M. (2020). Response to Ring et al.: In Silico Predictive Metagenomic Analyses Highlight Key Metabolic Pathways Impacted in the Hidradenitis Suppurativa Skin Microbiome. ***The Journal of Investigative Dermatology***, 140(7), 1476-1479. [Paper; Scopus Q: $\mathbf{Q}_1$]

11. Patel, V. A., Dunklebarger, M., **Banerjee, K.**, Shokri, T., Zhan, X., & Isildak, H. (2020). Surgical Management of Vestibular Schwannoma: Practice Pattern Analysis via NSQIP. ***Annals of Otology, Rhinology & Laryngology***, 129(3), 230-237. [Paper; Scopus Q: $\mathbf{Q}_2$]

12. Schneider, A. M., Cook, L. C., Zhan, X., **Banerjee, K.**, Cong, Z., Imamura-Kawasawa, Y., ... & Nelson, A. M. (2020). Loss of Skin Microbial Diversity and Alteration of Bacterial Metabolic Function in Hidradenitis Suppurativa. ***The Journal of Investigative Dermatology***, 140(3), 716-720. [Paper; Scopus Q: $\mathbf{Q}_1$]

13. **Banerjee, K.**, Zhao, N., Srinivasan, A., Xue, L., Hicks, S. D., Middleton, F. A., ... & Zhan, X. (2019). An adaptive multivariate two-sample test with application to microbiome differential abundance analysis. ***Frontiers in Genetics***, 10, 350. [Paper; R software; Scopus Q: $\mathbf{Q}_2$]

    [Selected as one of the top five finalists in the student paper competition at the IISA (International Indian Statistical Association) 2019 conference.]

14. Zeng, Y., Zhu, X., Chen, C., **Banerjee, K.**, Sun, L., Yu, W., ... & Wu, R. (2019). A unified DNA sequence and non-DNA sequence mapping model of complex traits. ***The Plant Journal***, 99(4), 784-795. [Paper; Scopus Q: $\mathbf{Q}_1$]

Manuscripts in Preparation

1. High definition spatial transcriptomic analysis of breast cancers with BRCA mutations.
2. Integrative spatial metabolomic and transcriptomic profiling of renal cancer progression.
3. The ADAM15/c-Src signaling axis promotes the invasive progression of human bladder cancer.
4. Spatial transcriptomic profiling of TRIM29-driven immune evasion in bladder cancer.

Abstracts

1. Gong, H., The, S., **Banerjee, K.**, Robinson, T., Monda, S., Salami, S. S., Keller, E. T., & May, A. (2026). Abstract A005: Biologic insights from spatial analysis of sarcomatoid renal cell carcinoma metastasis to inform therapeutic strategies. ***Cancer Research***, 86(5_Supplement_2), A005. [Link]
2. Nolan, Z. T., **Banerjee, K.**, Cong, Z., Gettle, S., Longenecker, A., Zhan, X., ... & Nelson, A. (2021). Isotretinoin disrupts skin microbiome composition and metabolic function after 20 weeks of therapy. ***Journal of Investigative Dermatology***, 141(5), S39. [Link]
3. Peterson, B., **Banerjee, K.**, Zhan, X., & Foy, A. (2019). Patients With Prior Myocardial Infarction or Prior Percutaneous Coronary Intervention Have Increased Perioperative Mortality and Adverse Cardiac Events Following Bariatric Surgery. ***Circulation***, 140(Suppl_1), A12242-A12242. [Link]

Commentary on Website

- **Banerjee, K.**, & Keller, E. T. (2025). Cellular annotation: the keystone of spatial analysis. ***BioTechniques***. [Link]

Presentations

- "A Robust Computational Framework for Identifying Copy Number Variations in Spatially Resolved Transcriptomics"
  Invited, Innovation Across Disciplines Seminar Series (April, 2026), Department of Mathematics and Statistics, South Dakota State University, USA.
- "Robust Inference of Copy Number Variations in Spatial Transcriptomics"
  Joint Statistical Meetings (JSM) 2025, USA.
- "An adaptive and powerful multivariate test for microbiome association analysis via feature selection"

- Joint Statistical Meetings (JSM) 2021, USA.
  (**Invited as a paper award winner from the Biometrics section**).
- International Indian Statistical Association (IISA) conference 2021, India.
  (**Winner of the Student Paper Competition**).

- "An adaptive test for microbiome association studies via feature selection"
  Eastern North American Region (ENAR) spring meeting 2021, USA.
- "An adaptive multivariate two-sample test with application to microbiome differential abundance analysis"
  - Joint Statistical Meetings (JSM) 2020, USA.
  - International Indian Statistical Association (IISA) conference 2019, India.
    (Invited as a finalist of the Student Paper Competition).

Professional Experience

Division of Biostatistics and Bioinformatics, Pennsylvania State University

- Teaching Assistant for Graduate-level courses
  - PHS 523: **Multivariate Analysis** (Fall 2020, online)
    Enrollment: Ph.D. students in Biostatistics.
  - PHS 522: **Multivariate Biostatistics** (Summer 2020, online)
    Enrollment: Ph.D., DrPH, MPH, and MD students from various disciplines.
  - PHS 520: **Principles of Biostatistics** (Fall 2019, online; Fall 2018, in-person)
    Enrollment: Ph.D., DrPH, and MD students from various disciplines.
- Research Assistant (Fall 2017 - Spring 2021)

Service

- Session Chair, Senior Ph.D. Student Research Showcase 2023 and 2024, Department of Biostatistics, University of Michigan.
- Judge, JSM 2023 Paper Award Committee, Biometrics section, American Statistical Association.
- Chair, Contributed Session: "Statistical Methods for Large-scale Omics Data", ENAR 2021.

Reviewing Activities

Nature Biotechnology, Frontiers in Genetics, and mSystems.

Computational Skills

R, Python, and SAS.

Awards/Honors

- *JSM (Joint Statistical Meetings) 2021 Paper Award* from the Biometrics Section of the American Statistical Association. [Twitter] [Penn State News]
- *Best Student Paper Award in Applied Statistics* at the IISA (International Indian Statistical Association) 2021 conference. [Announcement]

- *Graduate Alumni Society Award* (Penn State College of Medicine, 2020) for outstanding academic and research achievements and contributions to the campus community. [Penn State News]
- *Debesh-Kamal Scholarship* (INR 1 lakh, 2017 - 2018) for the purpose of higher education/research abroad from the Ramakrishna Mission Institute of Culture, Kolkata, India.
- Ranked 2^{nd} in the 3-year Bachelor of Science (B.Sc.) Honours examination (2015), University of Calcutta, India.
  - *Hasi Ganguly Memorial Prize* (Asutosh College, 2014 - 2015) for securing the highest marks in the B.Sc. final examination in Statistics Honours.
  - *Kesab Lal Ganguly Memorial Prize* (Asutosh College, 2013 - 2014) for securing the highest marks in B.Sc. Statistics Honours Part-II examination.
- Selected as a member of the 5^{th} delegation from India in the *Japan-East Asia Network of Exchange for Students and Youths Programme (JENESYS)* conducted by the Japan International Cooperation Center (June, 2008).

Contact Information for References

- **Evan T. Keller**
  Professor of Urology and Pathology,
  Director, Single Cell Spatial Analysis Program,
  University of Michigan, USA.
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- **Seth Pantanelli**
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